

Let $A = \{-5, -4, -3, -2, -1, 0, 1, 2, 3, 4, 5\}$

$$R_1 = \{(a, b) : a \leq b\}$$

$$\forall a \in A : (a, a) \in R_1 \quad \therefore \text{Reflexive}$$

$$\forall a, b \in A : (a, b) \in R_1 \Rightarrow (b, a) \in R_1 \quad \therefore \text{Not symmetric}$$

$$\forall a, b \in A : (a, b) \in R_1 \wedge (b, a) \in R_1 \Rightarrow a = b \quad \therefore \text{Antisymmetric}$$

$$\forall a, b, c \in A : (a, b) \in R_1 \wedge (b, c) \in R_1 \Rightarrow (a, c) \in R_1 \quad \therefore \text{Transitive}$$

b) Partial Order, b/c of def.

c) Hasse Diagram



$$R_2 = \{(a, b) : a^3 = b^3\}$$

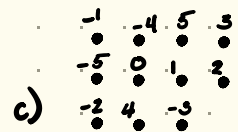
$$\forall a \in A : (a, a) \in R_2 \quad \therefore \text{Reflexive}$$

b) Equivalence & Partial Order

$$\forall a, b \in A : (a, b) \in R_2 \Rightarrow (b, a) \in R_2 \quad \therefore \text{Symmetric}$$

$$\forall a, b \in A : (a, b) \in R_2 \wedge (b, a) \in R_2 \Rightarrow a = b \quad \therefore \text{Antisymmetric}$$

$$\forall a, b, c \in A : (a, b) \in R_2 \wedge (b, c) \in R_2 \Rightarrow (a, c) \in R_2 \quad \therefore \text{Transitive}$$



$$R_3 = \{(a, b) : a \times b \leq 0\}$$

$$\forall a \in A : a \times a \leq 0 \quad \text{Proof by example: } 1 \times 1 \leq 0 = 1 \leq 0 \quad \therefore \text{NOT Reflexive}$$

$$\forall a, b \in A : a \times b \leq 0 \Rightarrow b \times a \leq 0 \quad \therefore \text{Symmetric}$$

$$\forall a, b \in A : a \times b \leq 0 \wedge b \times a \leq 0 \Rightarrow a = b \quad \therefore \text{NOT Antisymmetric}$$

Proof by example: $a=1, b=-1$: $1 \times -1 \leq 0 \wedge -1 \times 1 \leq 0 \Rightarrow 1 = -1$
 $-1 \leq 0 \wedge -1 \leq 0 \Rightarrow 1 = -1$
 $1 \neq -1$

$$\forall a, b, c \in A : a \times b \leq 0 \wedge b \times c \leq 0 \Rightarrow a \times c \leq 0 \quad \text{NOT Transitive}$$

Proof by example: $a=-2, b=1, c=-1$
 $-2 \times 1 \leq 0 \wedge 1 \times -1 \leq 0 \Rightarrow -2 \leq 0$
 $-2 \leq 0 \wedge -1 \leq 0 \Rightarrow -2 \leq 0$

2- Let R be a relation on $\mathbb{N}$ such that $x R y : 3 | x - y$.

Prove that R is an equivalence relation.

Identify the equivalence classes of R .

(8 points)

$$\forall x \in \mathbb{N} : 3 | (x - x)$$

Reflexive

$$x - x = 0, 0 = 3 \times 0;$$

$$\forall x, y \in \mathbb{N} : \underbrace{3 | x - y}_{3K} \Rightarrow 3 | y - x \quad \text{Symmetric}$$

$$\Rightarrow 3(-K)$$

$$\forall x, y, z \in \mathbb{N} : \underbrace{3 | x - y}_{3K} \wedge \underbrace{3 | y - z}_{3L} \Rightarrow 3 | x - z \quad \text{Transitive}$$

$$K, L \in \mathbb{Z} \quad 3(K+L)$$

Equivalence Classes

Class

$$[0] \quad x \in \mathbb{N} : x = 3K \quad 0, 3, 6, 9, 12$$

$$[1] \quad x \in \mathbb{N} : x = 3K + 1 \quad 1, 4, 7, 10, 13$$

$$[2] \quad x \in \mathbb{N} : x = 3K + 2 \quad 2, 5, 8, 11, 14$$

4- Using set identities and the fact that $x - y = x \cap \bar{y}$ prove that $(A - B) - (B - C) = A - B$. (8 points)

$$1. (A - B) - (B - C) \quad \square$$

$$2. (A - B) \cap \overline{(B - C)} \quad 1, \text{ def of } -$$

$$3. (A \cap \bar{B}) \cap (\overline{B \cap \bar{C}}) \quad 2, \text{ def of } -$$

$$4. (A \cap \bar{B}) \cap (\bar{B} \cup \bar{C}) \quad 3, \text{ d.m}$$

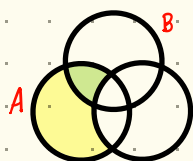
$$5. \underbrace{(A \cap \bar{B})}_X \cap (\bar{B} \cup C) \quad 4, \text{ d.n}$$

Let $X = A \cap \bar{B}$

$$6. (X \cap \bar{B}) \cup (X \cap C) \quad 5, \text{ distributive}$$

$$7. (A \cap \bar{B} \cap \bar{B}) \cup (A \cap \bar{B} \cap C) \quad \text{independent}$$

$$A \cap \bar{B} \stackrel{8.}{=} (A \cap \bar{B}) \cup (A \cap \bar{B} \cap C) \quad 6. \quad \text{Since } A \cap \bar{B} \cap C \text{ is a subset of } A \cap \bar{B}, \text{ the union between them is } A \cap \bar{B}$$



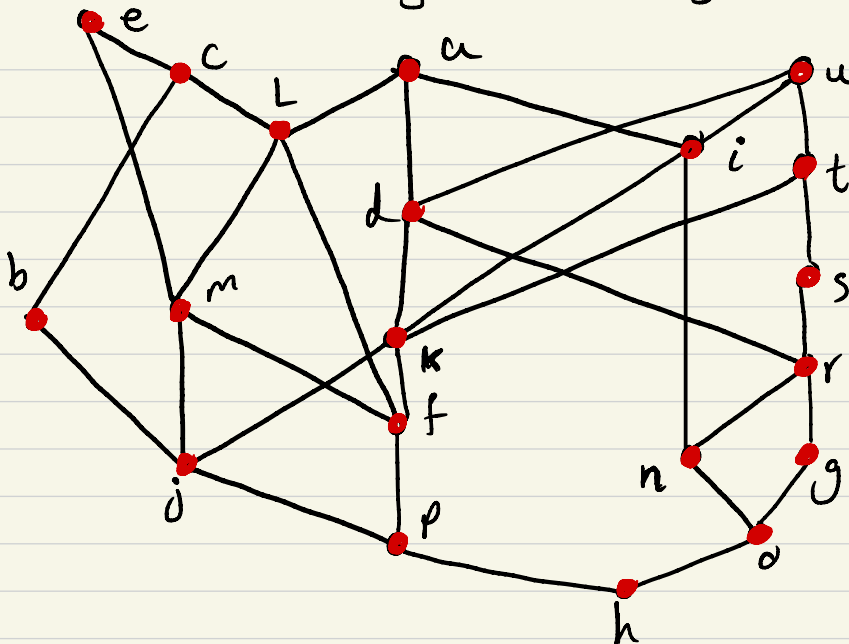
$A \cap \bar{B}$ from Absorption

$$9. A \cap \bar{B}$$

Here is a diagram to visualize the idea

3- Find the following items in the given Hasse diagram.

(10 points)



a) $ub(\{m, k, t\}) = \{\}$

b) $lb(\{c, d, t\}) = \{j, f, p, h\}$

c) $glb(\{u, k, m\}) = \{f\}$

$$d) \text{lub}(\{b, f\}) = \{c\}$$

e) $glb(\{p, k\}) = \{p\}$ ← Lecture 11 shows that you can include itself

$$f) \text{ glb}(\underbrace{\text{lub}(\{n, k, j\})}_{\{d, t, i, u, a\}}) = \{k\}$$

Question

Lecture 11 shows that $\sqcup$, $\sqcap$, $\sqcup$, and $\sqcap$ can include themselves, is this true? Prof. said more don't

Prof said one or more don't include themselves, but which?

$S = \{5, 6\}$ $T = \{2, 3, 4\}$
 $ub(S) = \{10, 12, 13\}$ $ub(T) = \{9, 8, 11, 9\}$
 $lub(S) = 10$ $lub(T) = 4$
 $lb(S) = \{3, 1\}$ $lb(T) = \{1\}$
 $gib(S) = 3$ $gib(T) = 1$
 $alb(\{3, 4, 5\}) = 3$ $alb(\{4, 5, 6, 7\}) = ?$

5 - Using set-builder notation prove that

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$$

$$\Leftrightarrow 1. x \in (A \cap (B \cup C)) \quad \text{1. def of s.b.n}$$

$$\Leftrightarrow 2. x \in A \wedge x \in (B \cup C) \quad \text{1, def of } \cap$$

$$\Leftrightarrow 3. x \in A \wedge (x \in B \vee x \in C) \quad \text{2, def of } \vee$$

$$\Leftrightarrow 4. (x \in A \wedge x \in B) \vee (x \in A \wedge x \in C) \quad \text{3, distributive}$$

$$\Leftrightarrow 5. x \in (A \cap B) \vee x \in (A \cap C) \quad \text{4. def of } \cap$$

$$\Leftrightarrow 6. x \in (A \cap B) \cup (A \cap C) \quad \text{5. def of } \cup$$

$$\Leftrightarrow 7. (A \cap B) \cup (A \cap C) \quad \text{6. def of s.b.n}$$